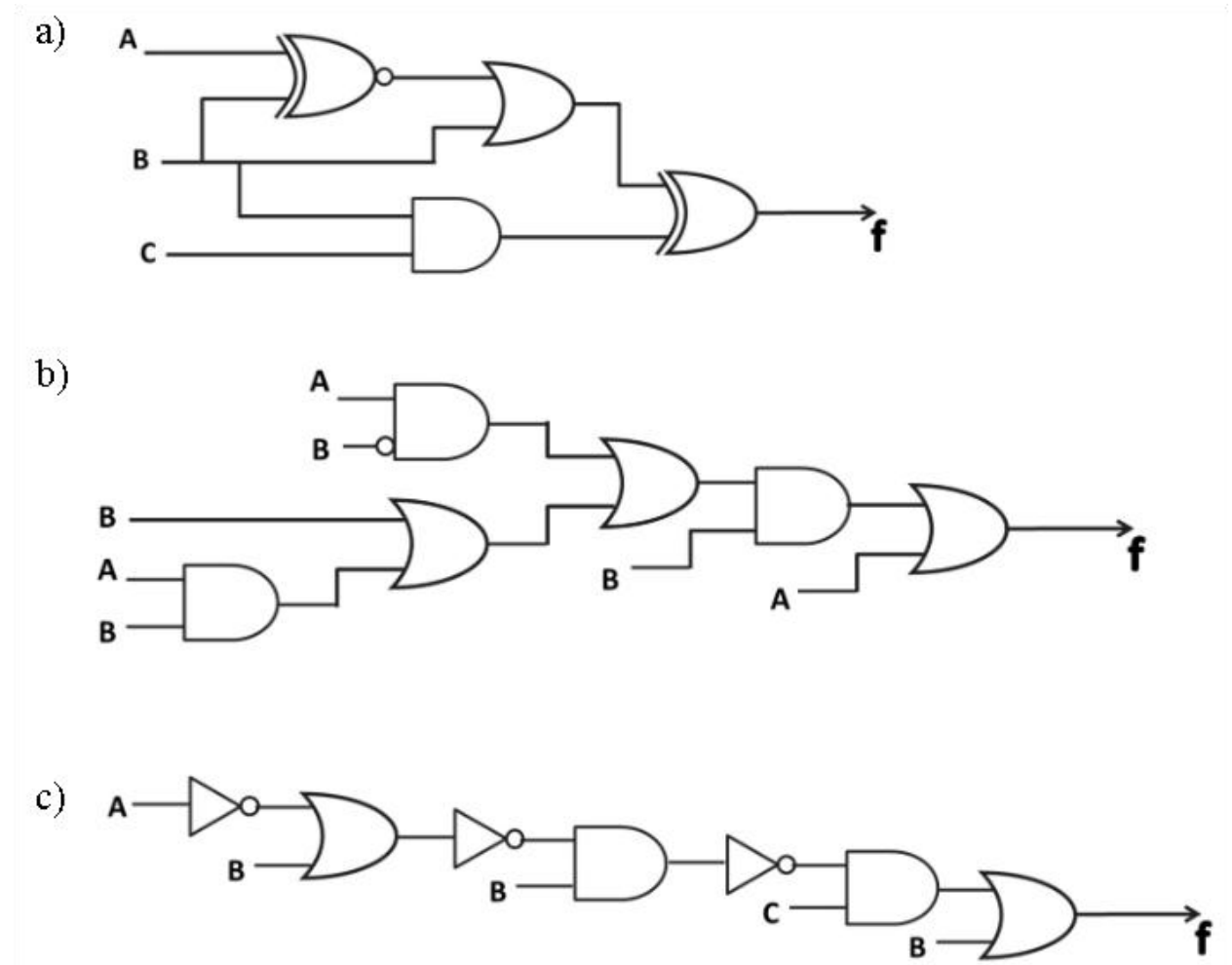


Boolean Algebra- Assignment

1. Derive the output expression 'f' for the below circuits and draw the simplified circuits.



2. Minimize the following Boolean Expressions using Boolean postulates.

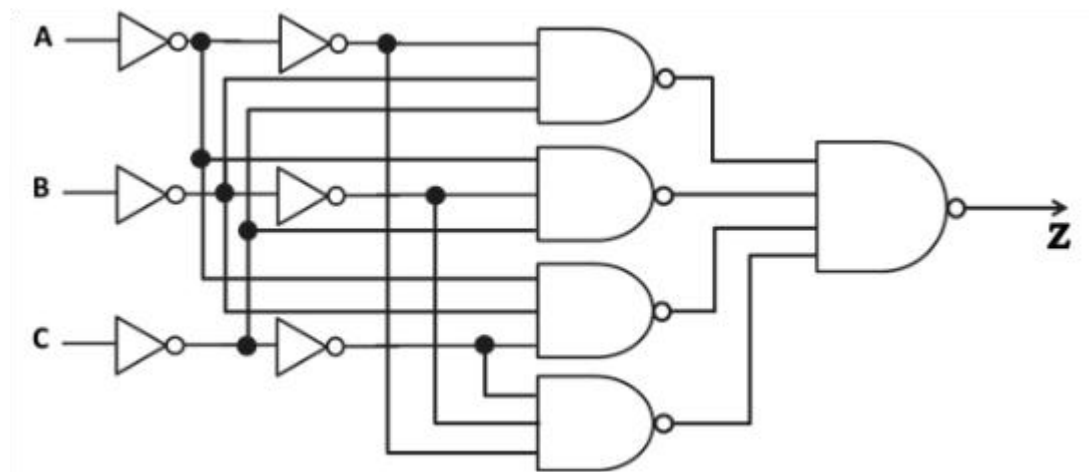
a) $\bar{A}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}\bar{C} + \bar{A}\bar{C}\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

d) $\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$

3. Which single gate can represent the functionality of given circuit?



4. Minimize the following Boolean Expressions using K - map

a) $f(a, b, c, d) = \sum_m(4, 6, 8, 10, 11, 12, 15) + \sum_d(3, 5, 7, 9)$

b) $f(a, b, c, d) = \prod_M(0, 2, 4, 6, 7, 9) \cdot \prod_D(10, 11)$